per requirement. His responsibilities include managing the product after deployment along with the content security policies (CSP) to make sure the product is working the way it should. He also needs to coordinate with R&D department and other teams within the organisation to make sure that the right solution is provided through the existing electronics of their product to make it work seamlessly.

Legal advisor: Conformance to numerous legal and regulatory requirements is most important for any IoT product or solution. Constant change in data storage rules of different countries makes it even more necessary to have a legal team that cross-checks all the related laws and requirements before a product is placed in the market.

IoT enabler: As the name suggests, an IoT enabler ensures delivery of time-bound end-to-end solution as per requirement. Sunil David, regional director of IoT, AT&T, says, "It is a very critical role because many times you find projects stuck in the product cycle as they need somebody who is an IoT champion within the organisation who can work inside the organisation."

Lifecycle manager: He manages the solution once it is implemented and makes sure that the solution keeps working well for the customer.

Security officer: Customers are usually afraid of the threats of their IoT system's data being leaked to malicious parties. These threats pose a major need for the data to be properly secured as it may contain sensitive information about the user. According to David, "With a huge shortage in India for cybersecurity professionals, individuals with a career in hardware or software security are most needed."

Data scientists: With complexity in dealing with huge volumes of ever-flowing data, the need for data scientists cannot be overestimated for an organisation. Their role is primarily to prepare meaningful and useful data from the huge volume of data being generated.

Gaps in the industry and skill-sets to develop

Lakshmi Mittra, VP & head, Clover Academy, says, "The diversity in the field of IoT eliminates the need for an individual to be from a specific background." For roles such as system integrator, solution architect, or IoT provider, candidates need to have the skill-set of dealing with sales and businesses as well, not just the technical aspect, because that is where the pain point of the customer lies. No doubt technical skills are what drive the IoT innovation and development, but in today's era of different domains in an organisation intertwining and collaborating, having a wide range of soft skills is just as important, if not more.

Given how IoT solves the day-to-day problems of its customers and aims to make life easier, having a flair for innovative thinking and creative abilities to look at the challenges and come up with the right solution is also essential. Additionally, having a curious mindset that searches for problems and solutions in everyday life is what sets them apart. The next gap to bridge would be the soft skills and need for creativity. "I have seen that people who know how to design innovative mechanical products at the right design parameters, cost and consideration, especially complicated ones like wearables, are mostly in demand because it is a very important skill to have," adds Solanki.

One of the most important skill-sets an aspiring candidate needs to possess is good communication skills. IoT is a domain that thrives on team effort. There are way too many aspects in this game for a single player to master, and it is imperative to have the right team from a product standpoint, putting the complete situation together so that the product is good. Good interpersonal skills are necessary because it requires collaboration and communication. Communication skills can be termed as another gap that has to be bridged.

"I think someone who can articulate and communicate well is very important. Even is at the junior level, people need to build other skill-sets in the industry and not just be technology experts. I think what can take you farther and help you climb up the ladder of career faster is that if you are also good in soft skills," adds Mittra.

Building your career as a fresher

Our education system lags far behind in imparting students with industry-related skills and knowledge. So, interested candidates need to start training themselves through internships, workshops, and courses as it is futile to depend on the system. Students and freshers from such an environment are often worried about how to make it big in the IoT space. Students of Tier 2 or Tier 3 cities are especially confused about how they could gain an edge over their peers from elite institutes.

David notes, "One of the gaps that I see often in such candidates is the lack of industry touch in comparison to the candidates from metro cities. They really don't know the practical aspects of what they are doing, or how they are going to shape their career." Hence, expanding their area of interest and having knowledge of the business aspect of the product or solution is very important. This aspect is something that young professionals tend to ignore as they strive to use their technical expertise alone.

"Apart from the technical skills, I think it is important to have the willingness to learn. The whole attitude is so important," David adds. To take on higher challenges of the domain for any student or fresher, learning and upskilling needs should be the highest priority. For instance, becoming a better coder, learning the marketing aspects of IoT, or familiarising yourself with cloud services like AWS or Microsoft Azure is now easier with the help of the Internet.